

FocusBoard

Assignments

*10 Challenges to Extend FocusBoard
and Sharpen Your Skills*

AI Pro | Codebasics

Pick what excites you, skip what does not. Each assignment uses the same flow you learned in the build guide: Scope → PRD → tasks.md → phased build → test.

Quick Glance: All Assignments

#	Assignment	Tier
1	Daily Motivational Quote	Easy
2	Daily Focus Streak Counter	Easy
3	Dark Mode / Light Mode Toggle	Easy
4	Pomodoro Timer Widget	Medium
5	Wallpaper Rotation	Medium
6	Task Categories with Colored Tags	Medium
7	Focus Mode Schedule	Medium
8	Daily Focus Time KPI	Hard
9	Analytics Dashboard	Hard
10	Build a Different Chrome Extension Using the Flow	Capstone

How to Use This Document

You have a working FocusBoard. Now it is time to make it yours.

Each assignment below is a challenge. Not a tutorial. The flow is the same as the main guide:

1. Read the assignment and decide what you want to build
2. Write your scope in plain English
3. Feed it to Claude or ChatGPT and generate a fresh PRD (or update the existing one)
4. Feed that into Antigravity and generate tasks.md
5. Build phase by phase
6. Test in Chrome

FocusBoard Assignments

You can do these in any order. You can skip any you do not care about. The Capstone at the end is the real goal: applying this flow to a project of your own.

How assignments are organized

Tier	What it means	Number of assignments
Easy	Extend existing features. Low risk of breaking things.	3
Medium	New features that require planning.	4
Hard	Significant additions. Full flow required.	2
Capstone	Apply the flow to a brand new project of your own	1

Format of each assignment

Field	What it tells you
What you are building	A clear 1 to 2 sentence description
Why this is interesting	What makes it worth doing
New concepts you will learn	2 to 3 skills this assignment teaches
Hints for your PRD	Things to make sure you include when writing scope
Success criteria	How to know you are done

Easy Tier

Pick any one of these to get comfortable with the flow again. None of them require rebuilding existing features.

Assignment 1: Daily Motivational Quote

EASY

What you are building

A short motivational quote that appears below your focus message. A new quote rotates in each day. Users can build the quote list themselves or you can ship 30 to 50 defaults.

Why this is interesting

This is the gentlest possible extension. A small visible feature that teaches you the whole flow end to end without touching anything risky. Good warm-up.

New concepts you will learn

- Storing and reading from an array of data
- Date-based indexing (pick the same quote for the same day)
- Adding a new visual element to an existing layout without breaking it

Hints for your PRD

- Where does the quote appear? Below the focus message? Above it? In a side corner?
- Is the quote list hardcoded or user-editable from Settings?
- What happens if you scroll through all quotes? Loop back, or random?

Success criteria

- A quote appears on the dashboard
 - The same quote shows for the entire day and changes the next day
 - The layout still looks clean (the quote does not crowd the existing elements)
-

Assignment 2: Daily Focus Streak Counter

EASY

What you are building

A small counter somewhere on the dashboard that shows how many days in a row you have set a focus. Something like "12 day streak". If you miss a day, the streak resets to 0.

Why this is interesting

Streaks are one of the most effective motivation mechanics in product design (Duolingo built a company on this). Seeing the number grow makes setting your daily focus addictive.

New concepts you will learn

- Tracking state across days, not just within a day
- Date math (what counts as yesterday versus today versus 2 days ago)
- Persistent counters with conditional reset logic

Hints for your PRD

- Define exactly what breaks a streak: missing a day, or just not setting a focus?
- Where does the counter live visually?
- Do you want a longest streak ever stat too, for when the current one resets?

Success criteria

- Set a focus today, counter shows 1
- Set a focus tomorrow, counter shows 2
- Skip a day, counter resets to 0 the day after
- Counter persists across browser restarts

Assignment 3: Dark Mode / Light Mode Toggle

EASY**What you are building**

A theme toggle in Settings that switches the entire dashboard between a dark theme and a light theme. The overlays, text colors, and task panel all adapt.

Why this is interesting

Dark mode is a universal expectation now. This teaches you how to structure a design system using CSS variables, which is a skill you will use in every future project.

New concepts you will learn

- CSS custom properties (variables) for theming
- Swapping themes without reloading the page
- Keeping visual consistency across multiple components

Hints for your PRD

- Decide: does the wallpaper itself change, or just the UI overlays?
- Should the theme auto-follow the system theme, or always manual?

- Where does the toggle live in Settings?

Success criteria

- Toggle the theme in Settings, all colors update instantly
 - Reload the tab, theme persists
 - Both themes are readable. No invisible text anywhere.
-

Medium Tier

These are real features that require planning. Write a proper PRD before you start.

Assignment 4: Pomodoro Timer Widget

MEDIUM**What you are building**

A Pomodoro timer panel on the dashboard. Click start, it counts down 25 minutes of focus time, then 5 minutes of break, then repeats. Shows a Chrome notification when each phase ends.

Why this is interesting

Pomodoros are the single most proven productivity technique. Having it built into the tab you open 50 times a day means no excuse not to use it.

New concepts you will learn

- Working with timers and intervals (and cleaning them up properly)
- Chrome notifications API and permissions
- Managing multi-state widgets (work to break to work, with pause and reset)

Hints for your PRD

- Where does the timer live? A separate panel, or integrated with the task list?
- Should the timer keep running if you close the tab? (Hint: this is harder than it sounds)
- Allow configurable work/break lengths, or stick to classic 25/5?
- What happens when a Pomodoro completes: notification, sound, visual celebration?

Success criteria

- Start a 25 minute timer, counts down visibly
- Chrome notification fires when it hits 0
- Break timer starts automatically (or with one click)

- You can pause, reset, and skip to the next phase

Assignment 5: Wallpaper Rotation

MEDIUM

What you are building

Upload multiple wallpapers in Settings (not just one). Choose to rotate them daily, on every new tab, or on a custom schedule.

Why this is interesting

Extends a feature that already exists. Good practice for going back and modifying something without breaking it. Plus, rotating wallpapers keep the dashboard feeling fresh.

New concepts you will learn

- Managing arrays of files instead of single values
- Storage quota planning (multiple images adds up fast)
- Random vs sequential selection logic

Hints for your PRD

- Cap the number of uploads (5 to 10 max) to stay within Chrome's storage limits
- Preview all uploaded wallpapers in Settings with delete buttons on each
- Let the user preview each wallpaper before committing to the rotation

Success criteria

- Upload 3 wallpapers, all 3 appear in a Settings gallery
- Set rotation to every new tab, each new tab shows a different wallpaper
- Delete a wallpaper, it is removed from rotation
- Image quality stays high (no compression)

Assignment 6: Task Categories with Colored Tags

MEDIUM

What you are building

Add a category to each task (Work, Personal, Learning, and custom ones users can create). Each category has a color. Filter chips at the top of the task list let you show only tasks from one category.

Why this is interesting

The simple task list from the original build is intentionally basic. This is the natural make it more powerful upgrade without turning it into Todoist.

New concepts you will learn

- Expanding your data model without breaking existing data (migration thinking)
- Filtering a list based on user input
- Designing consistent color systems across many items

Hints for your PRD

- Decide on default categories (Work, Personal, Learning) and allow custom ones
- Let users pick colors from a preset palette, not a full color picker (easier UX)
- Existing tasks: keep them uncategorized, or bulk-assign on first load?
- Filter UI: chips at the top? A dropdown? Both?

Success criteria

- Add a task and pick a category, it shows in the right color
- Create a new custom category with a color, it appears in the picker
- Click a category filter chip, only tasks in that category show
- All existing tasks from before still work

Assignment 7: Focus Mode Schedule**MEDIUM****What you are building**

In Settings, set hours when Focus Mode should auto-enable (e.g., 9 AM to 12 PM, 2 PM to 5 PM). During those hours, Focus Mode turns on by itself. Outside those hours, it turns off.

Why this is interesting

Removes the friction of remembering to toggle Focus Mode. If you know mornings are your deep work time, set the schedule once and forget it. The system protects you from yourself.

New concepts you will learn

- Chrome alarms API for scheduled tasks that work even when the browser is closed
- Handling time ranges and overlaps
- Respecting manual overrides (if the user toggles it off during a scheduled window, should it stay off?)

Hints for your PRD

- How many scheduled blocks should be supported? 1? 3? Unlimited?
- What happens on weekends? Separate schedule? Apply the same one? Off entirely?
- If the user manually toggles Focus Mode off during a scheduled window, does it re-enable?

FocusBoard Assignments

- Show a visual indicator on the dashboard that Focus Mode was auto-enabled

Success criteria

- Set a schedule: 9 AM to 12 PM
- At 9 AM, Focus Mode turns on automatically
- At 12 PM, Focus Mode turns off automatically
- The schedule works even after closing and reopening Chrome

Hard Tier

Significant additions. Treat these like real mini-projects. Write a proper PRD. Break into phases.

Assignment 8: Daily Focus Time KPI

HARD

What you are building

A KPI display at the top of the dashboard showing total time Focus Mode has been ON today. Something like "Focused today: 2h 45m". Every minute Focus Mode is active, the timer ticks up. Resets at midnight.

Why this is interesting

Turns focus into a measurable number you can see. Seeing 0h 00m at 3 PM is a wake-up call. Seeing 4h 30m feels like a win. Focus becomes a game you play against yourself.

New concepts you will learn

- Accumulating time across multiple Focus Mode sessions in a day
- Daily reset logic that works across midnight (and timezone quirks)
- Designing KPI visuals that feel motivating, not cluttering
- Tracking background state reliably in a service worker

Hints for your PRD

- Where does the KPI sit visually? Near the clock? Above the greeting? In a corner?
- What format: 2h 45m, 2:45, or 165 min?
- Do you want a weekly total too, or just daily?
- What happens if someone toggles Focus Mode on and off rapidly? Should each session count?
- Consider showing a small daily goal like Focused today: 2h 45m / 4h goal

Success criteria

- Turn on Focus Mode for 30 minutes, KPI shows 0h 30m
- Turn it off, then on again for another 15 minutes, KPI shows 0h 45m
- Reload the tab, KPI still shows 0h 45m
- At midnight, the counter resets to 0h 00m
- The KPI looks polished and belongs on the dashboard, not tacked on

Assignment 9: Analytics Dashboard

HARD**What you are building**

A new Analytics view (accessible from a button or Settings) that shows charts and stats about your FocusBoard usage. How many tasks you completed this week, how often Focus Mode was ON, which sites you tried to visit while it was on, your focus streak over time.

Why this is interesting

This is the most impressive thing you can build on this foundation. It turns FocusBoard from a tool into a personal productivity system. Also, charts are just fun to look at.

New concepts you will learn

- Event logging (recording WHAT happened and WHEN, in the background)
- Data aggregation (turning raw events into weekly or daily summaries)
- Chart libraries (Chart.js is the easiest for beginners)
- Building a second view inside a Chrome extension

Hints for your PRD

- Decide what events to log: task completed, focus set, focus mode toggled, blocked site attempt
- Think about storage carefully. Logging every single event will balloon. Aggregate nightly if needed.
- Pick 3 to 5 key metrics to show. Not 20. Less is more on a dashboard.
- Where does the Analytics view live? A new tab? A modal? A separate chrome-extension page?

Success criteria

- Complete a few tasks, toggle Focus Mode a few times, try to visit Instagram while it is on
- Open the Analytics view
- See clear charts showing the activity over the past 7 days
- Numbers match your actual usage (no phantom events)

Capstone

This is the real goal of the entire guide. Everything else was practice.

Assignment 10: Build a Different Chrome Extension Using the Flow

CAPSTONE

What you are building

A completely new Chrome extension for a different use case. Apply the full flow from the main guide: Scope → PRD → tasks.md → phased build → test. You keep the technical setup (Chrome extension, Manifest V3, chrome.storage) that is now familiar, but solve a different problem.

Why this is interesting

You did not learn FocusBoard. You learned a methodology, on a specific platform. This assignment proves you can run the methodology again and build something new from scratch.

Ideas to start with

Extension idea	What it does
Link Vault	Save any webpage with one click. Tag it, add a note, find it later. Replaces messy bookmarks.
Reading List	Save articles you want to read later. Mark as read. Track what you actually finish.
Quick Notes	A new tab page that is just a notepad. Everything you type auto-saves. Good for scratch thoughts.
Tab Limiter	Block yourself from opening more than 10 tabs at a time. When you hit the limit, it asks you to close some first.
Site Time Tracker	Track how much time you spend on each website. Weekly summary in a new tab.
Content Idea Capture	Hit a keyboard shortcut on any page to save a quote or screenshot as a future LinkedIn or Instagram idea.
Gratitude Log	New tab prompts 3 things you are grateful for today. Saves them. Shows a streak.

Pick one from the list, or come up with your own. The constraint: it must be a Chrome extension, and it must solve a real problem you have.

New concepts you will learn

- Applying a methodology without being told exactly what to build
- Defining scope for your own project (harder than following someone else's)
- Knowing when to push the PRD further vs. when it is ready to build
- Owning decisions that no one else has made for you

Hints for your PRD

- Start small. Really small. You can always extend later.
- Your first scope will be too ambitious. Cut it in half.
- Write the scope like you are describing it to a friend, not an engineer.
- Do not skip any stage of the flow. Even if you feel like it.

Success criteria

- You have a working Chrome extension that you built end to end
- You followed the Scope to PRD to tasks.md to phased build flow without skipping stages
- It solves a real problem for you personally
- You can explain to someone else how you built it, using the flow as the structure

Show Your Work to the World

You built something real. Do not let it sit on your laptop.

Post it on LinkedIn

Share what you built. A short post, a screen recording, a screenshot of your dashboard. Whatever feels natural. Talk about:

- What you built and why
- The flow you followed (*Scope* → *PRD* → *tasks.md* → *phased build* → *test*)
- What surprised you during the build
- What you want to build next

Tag these people so we can see it:

- [Hemanand Vadivel](#)
- [Naveen S](#)
- [Aditya Goel](#)
- [Codebasics](#)

Use hashtags like [#CBAIPro](#), [#Codebasics](#), [#AIPROs](#).

Share it on Discord

Drop a screenshot, a demo video, or the zip of your extension in the Codebasics AI Pro Discord. Your cohort will celebrate it. Other learners will borrow ideas from it. Some of the best discussions in the community come from people sharing what they built.

Why this matters

Shipping something is already a big deal. Most people never get past the planning stage. But sharing what you shipped is what turns a build into a story.

What sharing does	Why it matters
Forces you to articulate what you built	Makes your learning stick 10x harder
Shows you are someone who actually does the work	Opens doors to conversations, collabs, and opportunities
Inspires others in the cohort to ship their own	You become the reason someone else starts

What sharing does	Why it matters
Builds your public portfolio over time	Three builds in, this becomes a pattern people notice

The flow you learned is more valuable than any single thing you built with it.

Scope → PRD → tasks.md → Build in phases → Test incrementally

Apply it to everything. **Ship it. Share it.**

Happy building.